

Signal and System Design (IGS2137)

Jump Together, Fly Farther!



인하대학교 국제학부

Week 6 Lab

**ISE Department
Prof. Mehdi Pirahandeh**

Review Exercise:

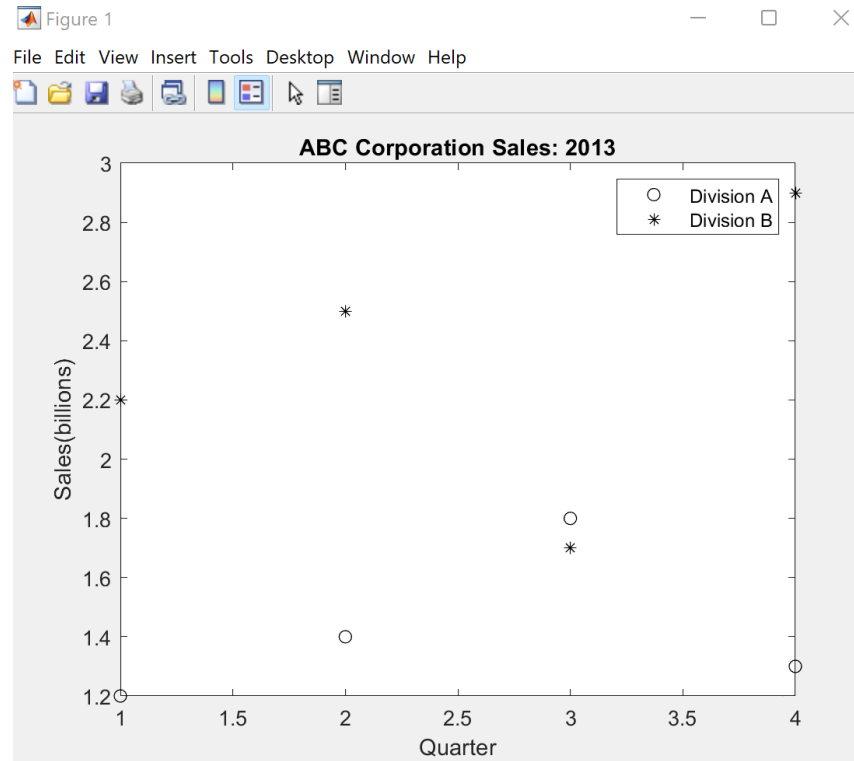
Write a script that will read ABC Corporation sales figures for divisions A and B from the given file in I-Class and plot it.

Review Exercise :

Write a script that will read ABC Corporation sales figures for divisions A and B from the given file in I-Class and plot it.

```
% Prompt the ABC Corporation sales figures
% Read sales figures from a file and plot it

load salesfigs.dat
asales = salesfigs(1,:);
bsales = salesfigs(2,:);
plot(asales,'ko')
hold on
plot(bsales,'k*')
xlabel('Quarter')
ylabel('Sales(billions)')
title('ABC Corporation Sales: 2013')
legend('Division A', 'Division B')
```



A sample plot for exercise 1:

The ABC Corporation sales figures

Exercise 1:

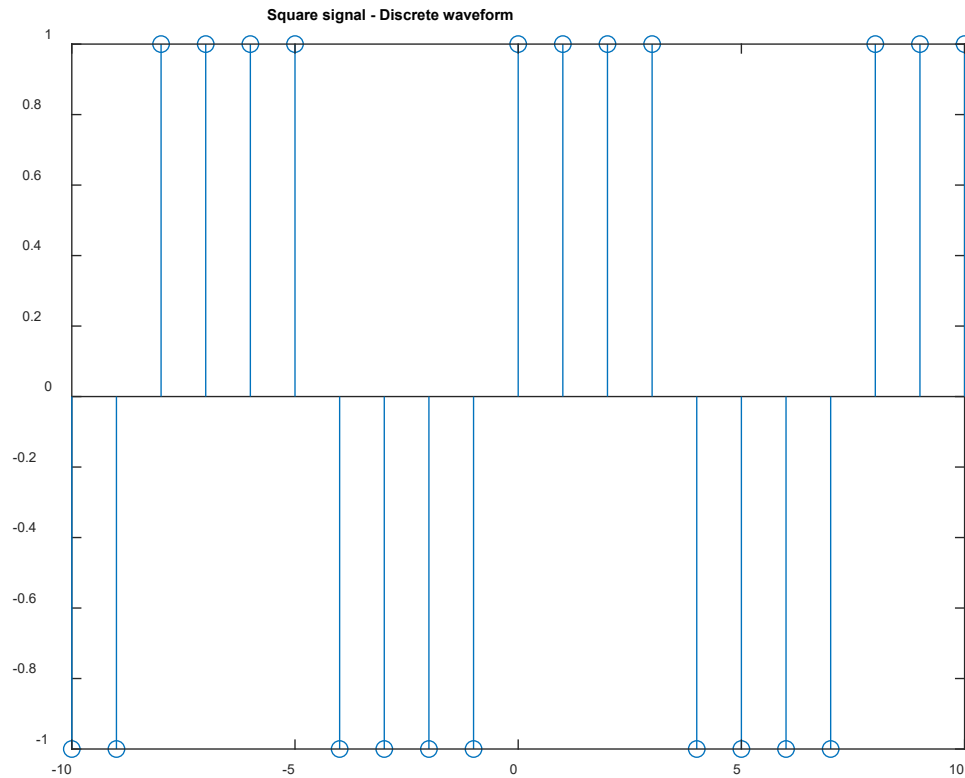
Write a script that will visualize discrete waveforms of periodic signals. Note that the time scale must be discrete and observe the changes.

Exercise 1:

Write a script that will visualize discrete waveforms of periodic signals. Note that the time scale must be discrete and observe the changes.

```
% Prompt the discrete waveforms of periodic signals
```

```
A=1  
omega = pi/4;  
n= -10:10;  
x = A* square(omega*n)  
figure, stem(n,x)  
title('Square signal - Discrete waveform ')
```

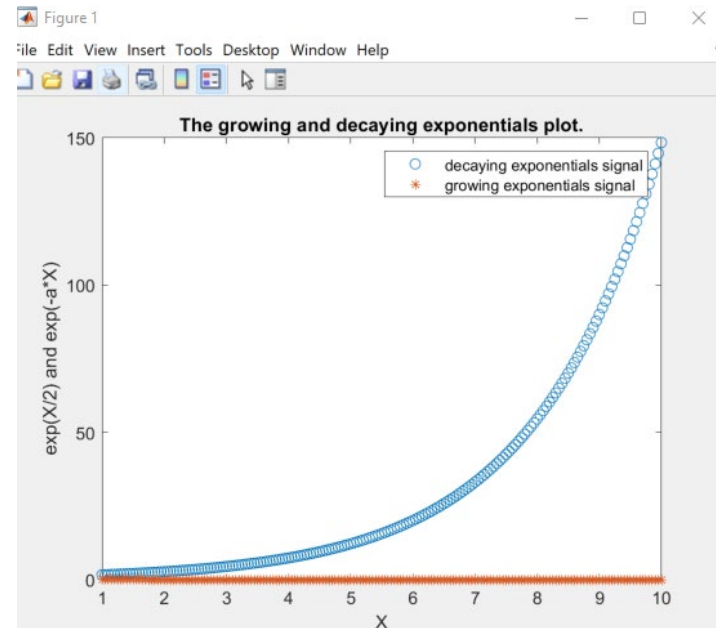
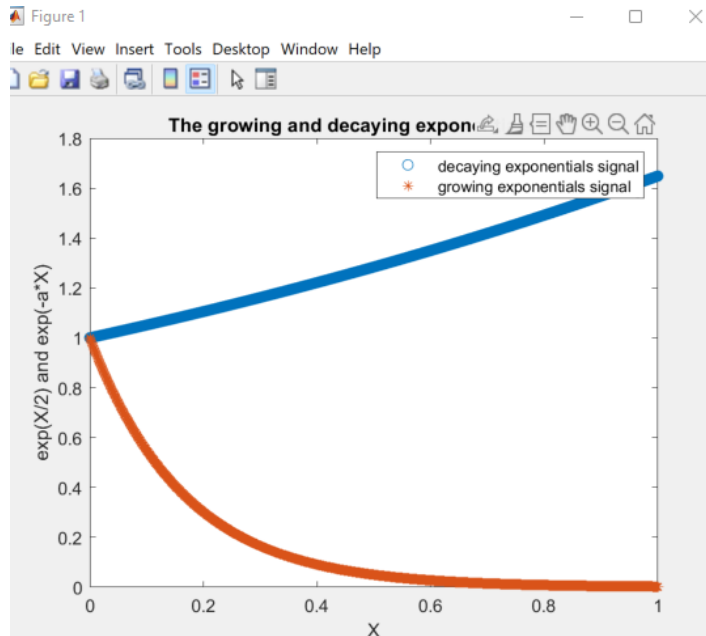


A sample plot for exercise 1:

The discrete waveforms of periodic signals.

Exercise 2:

Write a script that will visualize both the growing and decaying exponentials. Note that the time scale must be **continuous** and observe the changes.



A sample plot for exercise 2:

The growing and decaying exponentials signals plot.

Exercise 2:

Write a script that will visualize both the growing and decaying exponentials signals.

```
% Prompt the continuous the growing exponential signal
X = 0:0.001:1;
Y = exp(X/2);
plot(X,Y,'o')

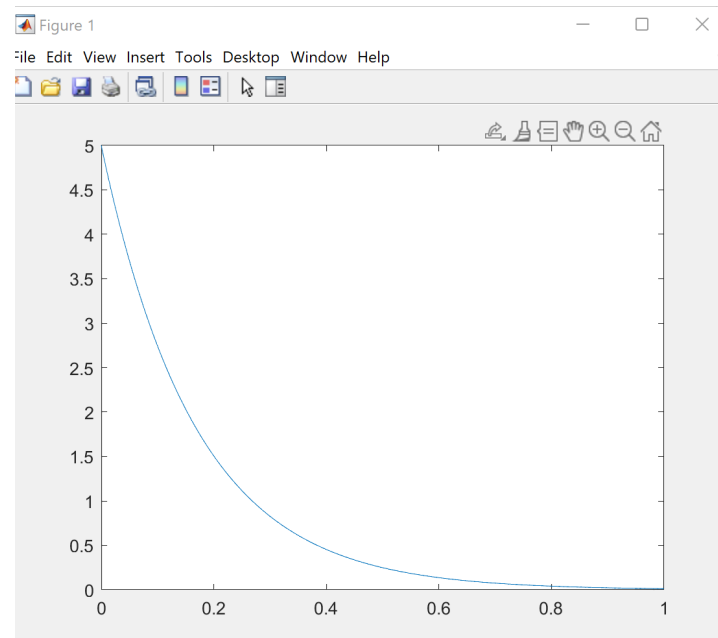
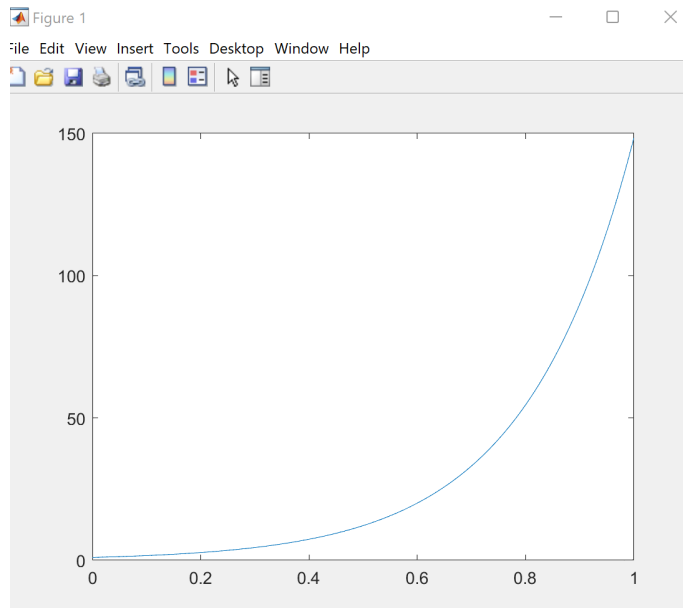
% Prompt the continuous decaying exponentials signal
hold on
a = 6;
Z = exp(-a*X);
plot(X,Z , '*');
legend('decaying exponentials signal', 'growing exponentials signal')
xlabel('X')
ylabel('exp(X/2) and exp(-a*X)')
title('The growing and decaying exponentials plot.')
```

Exercise 2:

Write a script that will visualize both the growing and decaying exponentials signals.

```
B = 1;  
a = 5;  
timing = 0:0.001:1;  
x = B*exp(a*timing);  
plot(timing, x)
```

```
B = 5;  
a = 6;  
timing = 0:0.001:1;  
x = B*exp(-a*timing);  
plot(timing, x)
```



A sample plot for exercise 2:

The growing and decaying exponentials signals plot.

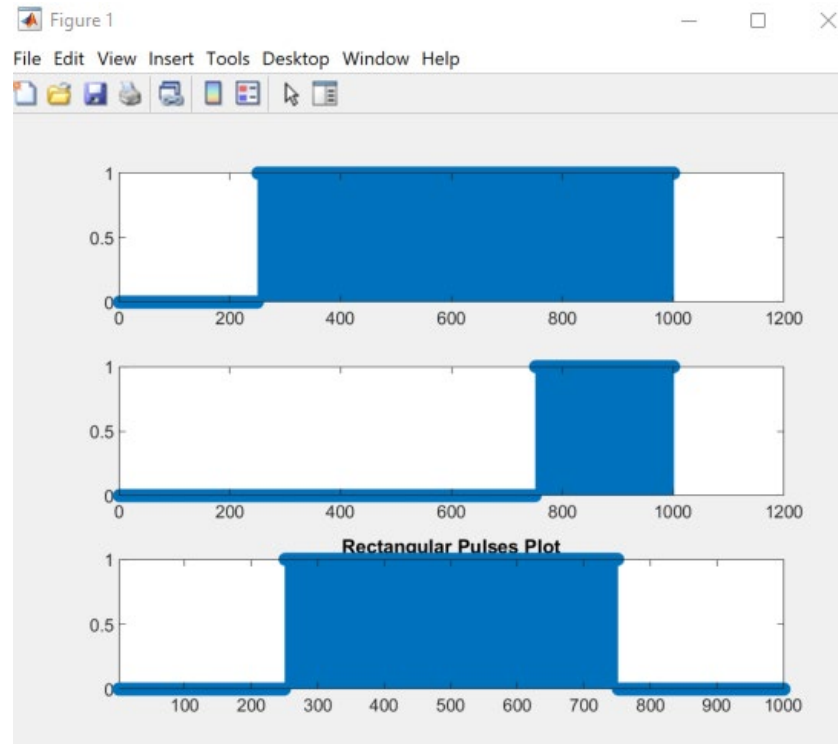
Exercise 3:

Write a script that will visualize three rectangular pulses of v_1 , v_2 and $v_1 - v_2$.

- 1000 samples indicate the interval of one second
- a gap of 2 ms.

Exercise 3:

```
t = -1:1/500:1
v1 = [zeros(1,250), ones(1,751)];
v2 = [zeros(1,751), ones(1,250)];
V = v1-v2
figure, subplot(3,1,1)
stem(v1)
    subplot(3,1,2)
stem(v2)
    subplot(3,1,3)
stem(V)
axis tight
title('Rectangular Pulses Plot')
```



A sample plot for exercise 3:

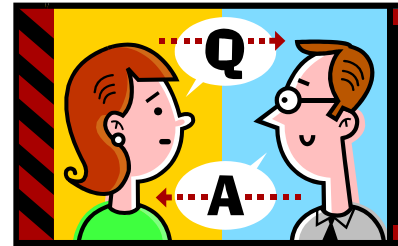
Exercise 4:

Write a script that will visualize ramp functions and merge them with a unit-amplitude step function is generated using ones (amplitude = 1). Note that the concatenation is performed using square braces in Matlab.

- $[A \ B]$ or $[A, B]$ $\rightarrow$ row-wise concatenation
- $[A; B]$ $\rightarrow$ column-wise concatenation
- Make sure that dimensions must comply accordingly

Brief break (if on schedule)

Q&A?



Prof. Mehdi Pirahandeh
E-mail : mehdi@inha.ac.kr